

RESEARCH ARTICLE

Comparative methods for determining MoCA, MMSE, and CDR cut-offs in Alzheimer's disease: insights from the ADNI study

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[#]Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

Abstract

INTRODUCTION: Accurate cut-off values for the Montreal Cognitive Assessment (MoCA), Mini-Mental State Examination (MMSE), and Clinical Dementia Rating (CDR) are essential for detecting and staging Alzheimer's disease-related cognitive impairment. Prior studies often relied on a single statistical method, potentially misaligning thresholds with clinical priorities.

METHODS: We analyzed cognitive assessments from 883 Alzheimer's Disease Neuroimaging Initiative participants classified as cognitively normal, mild cognitive impairment (MCI), or clinical impairment (CI). Optimal thresholds were identified using Euclidean distance, Index of Union, maximum product of sensitivity and specificity, and Youden's J Index. Sensitivity, specificity, and area under the curve were calculated.

RESULTS: For MCI, MoCA cut-offs were 24 (three methods) and 25 (Youden's J Index). For CI, MoCA cut-offs were 17 (Youden's J Index) and 18 (others). MMSE (28, 24) and CDR (0.5, 1) cut-offs were consistent.

DISCUSSION: MoCA thresholds vary by method, requiring context-specific selection, while MMSE and CDR cut-offs remain stable.

KEYWORDS

ADNI, Alzheimer's disease, CDR, cognitive impairment, cut-off methods, diagnostic thresholds, MMSE, MoCA

Highlights

- Four statistical methods were compared for cognitive test cut-offs in ADNI cohort.
- MoCA thresholds varied by method; MMSE and CDR cut-offs were consistent
- A MoCA cut-off of 25 favored sensitivity, 24 favored balanced accuracy.
- Findings guide context-specific screening for MCI and clinical impairment.
- Results highlight a need for validation in diverse, biomarker-confirmed cohorts.

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1 | BACKGROUND

Alzheimer's disease (AD) affects more than six million Americans and remains the leading cause of dementia worldwide, with prevalence projected to nearly double by 2060 due to population aging.¹ Early and accurate identification of cognitive impairment, particularly mild cognitive impairment (MCI) due to AD, is critical for timely intervention, prognosis, and enrollment in clinical trials of emerging therapies. Cognitive screening instruments such as the Montreal Cognitive Assessment (MoCA), the Mini-Mental State Examination (MMSE), and the Clinical Dementia Rating (CDR) scale are instrumental tools in both clinical practice and research.^{2–4} However, despite their widespread adoption, consensus on optimal cut-off values for these measures remains contentious, and reliance on single-method threshold determination risks misalignment with specific diagnostic priorities.⁵

Previous studies largely derived cut-offs for MoCA (commonly 26), MMSE (24 to 28), and CDR (0.5 for MCI) using single statistical criteria, most frequently Youden's J Index (YI) or expert consensus.^{6–8} Such approaches may inadequately balance sensitivity and specificity across diverse clinical and research settings, leading to variable diagnostic performance and hindered comparability of study findings. Moreover, the existing literature has paid limited attention to how different cut-off determination methods perform when applied to both MCI and CI within the same cohort.

To address these gaps, our study compares four statistical methods – Euclidean distance (ED), Index of Union (IU), maximum product of sensitivity and specificity (MSS), and YI – to identify optimal thresholds for MoCA, MMSE, and CDR in both MCI and CI using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). We evaluate the convergence and divergence of these methods and quantify diagnostic performance.

By integrating these evaluations with a focus on clinical implementation, this work aims to provide evidence-based guidance for selecting cognitive screening cut-offs tailored to specific diagnostic objectives and resource environments. The findings will inform clinicians and researchers on optimized threshold application, ultimately enhancing the precision and comparability of cognitive assessment in AD.

2 | METHODS

2.1 | Study design and participants

Data used in the preparation of this article were obtained from the ADNI database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of MCI and early AD.

RESEARCH IN CONTEXT

- Systematic review:** A comprehensive PubMed and conference abstract search were conducted to identify studies on cut-off determination for MoCA, MMSE, and CDR. While individual papers evaluated thresholds using single criteria (most commonly the Youden Index), no prior work directly compared multiple statistical approaches within the same well-characterized cohort.
- Interpretation:** By applying four complementary methods – Youden Index, ED, Index of Union, and MSS – to ADNI data, our study demonstrates consistent optimal thresholds for CDR and MMSE across diagnostic stages and highlights method-dependent variability in MoCA cut-offs. These findings extend existing literature by quantifying convergence and divergence among approaches and offering evidence-based guidance for threshold selection aligned with clinical priorities.
- Future directions:** Future research should (a) validate these multimethod thresholds in diverse, community-based cohorts; (b) assess the longitudinal stability and prognostic value of cut-offs; and (c) integrate optimized cognitive thresholds with emerging fluid and imaging biomarkers to refine diagnostic algorithms.

2.2 | Data collection

Inclusion criteria included all participants with three cognitive test results (MMSE, MoCA, and CDR) at their most recent evaluation and a gold-standard physician diagnosis (cognitively normal [CN], MCI, or CI) spanning the entire ADNI cohorts (ADNI 1, ADNI Go, ADNI 2, ADNI 3, and ADNI 4). No exclusion criteria were applied, aside from missing data, to maximize the number of eligible participants.

Alongside cognitive test results, demographic information (age, sex, education, ethnicity, and occupation) and anthropometric measurements (weight and height) were also collected. Medical history was also gathered across multiple domains, including psychiatric, neurological (excluding cognitive history), cardiovascular, respiratory, musculoskeletal, and endocrine-metabolic disorders. Family history of dementia was also collected. During the data-cleaning process, body mass index (BMI) was calculated from the recorded weight and height. BMI groups using standard thresholds (<18.5, 18.5 to 24.9, 25 to 29.9, and >30 kg/m²) were also populated.

2.3 | Cognitive spectrum definitions and clinician diagnostic procedure

This study included participants from all ADNI cohorts. A list of complete inclusion and exclusion criteria can be found in the ADNI

documentation.⁹ Below are brief definitions, specifically related to MMSE and CDR.

CN is composed of individuals who do not present significant evidence of cognitive impairment during their screening assessment with a MMSE score of 24 to 30 (inclusive) and a CDR of 0. The MCI cohort represents an intermediate stage in the progression to CI. These participants do not meet the diagnostic criteria for AD cohort but still present clinically relevant cognitive impairment with the same MMSE as CN but have a CDR of 0.5. ADNI enlists newly enrolled participants with early-stage AD that have a MMSE of 20 to 28 (inclusive) and a CDR of 0.5 or 1. Given the longitudinal nature of the cohort, participants from previous ADNI cohorts continue enrollment and, thus, disease progression. In this paper, CI refers to the entire spectrum of AD within the ADNI cohort, not just early-stage AD. ADNI also has screening criteria to exclude prospective participants in the early-stage AD arm whose symptoms likely arise from a non-AD etiology, such as severe cerebrovascular disease, frontotemporal dementia, or Parkinson's disease.

ADNI site physicians assigned diagnostic labels by integrating multiple sources of information available at the screening and baseline visits, following consensus guidelines based on the National Institute on Aging-Alzheimer's Association framework.¹⁰ Typical inputs included a structured clinical interview with the participant, collateral history from an informant, standardized assessments of activities of daily living and functional status, results from a comprehensive neuropsychological battery (in addition to the screening instruments analyzed here), a focused neurological examination, structural neuroimaging (primarily MRI), and routine laboratory testing to exclude other causes of cognitive decline. Diagnostic adjudication followed consensus guidelines at each ADNI phase and relied on clinician judgment to synthesize these data. Because ADNI enrollment criteria permit overlapping score ranges across diagnostic groups (e.g., MMSE of 20 to 28 are allowable in the dementia arm), clinician labels are derived synthesis rather than a single-test result. Thus, it is treated it as a pragmatic clinical gold standard while acknowledging that site-level practices and the relative weight given to individual tests may vary.

2.4 | Outcome measures

The primary outcome is to assess the four statistical methods of determining cut-off values for each cognitive test. Each method aims to maximize cut-offs differently. The YI maximizes the vertical distance of the receiver operating characteristic (ROC) curve from the diagonal (chance line) to capture the optimal balance between sensitivity and false positives. The ED minimizes the distance between the ROC curve and the ideal point (0,1). The IU method minimizes misclassification by balancing sensitivity and specificity relative to the area under the curve (AUC). Lastly, the MSS method identifies the threshold at which the product of sensitivity and specificity is maximized.

2.5 | Statistical analysis

Statistical analyses were performed using Python (version 3.13) with the scikit-learn package (version 1.7), Pandas (version 2.2), and NumPy (version 2.3). Sensitivity and specificity for each cognitive test are computed using 2×2 contingency tables. ROC curves were generated to evaluate the overall discriminative performance of each screening tool, and pairwise comparisons among the tests were conducted. Between-group comparisons of demographic and clinical characteristics were performed to identify features that differed across diagnostic categories. Continuous variables (age, years of education, and cognitive test scores) were assessed for normality and compared using one-way analysis of variance (ANOVA). Non-normally distributed continuous variables were compared using the Kruskal-Wallis test. Categorical variables (sex, race/ethnicity, and occupation categories, family history of dementia, and presence of comorbid conditions) were compared using Pearson's chi-squared (χ^2) test. Measures of medical comorbidity were summarized using available chronicity and severity indices and compared across groups using ANOVA or Kruskal-Wallis as appropriate. A two-sided alpha of 0.05 was used to define statistical significance.

3 | RESULTS

Descriptives (Supplemental Table A):

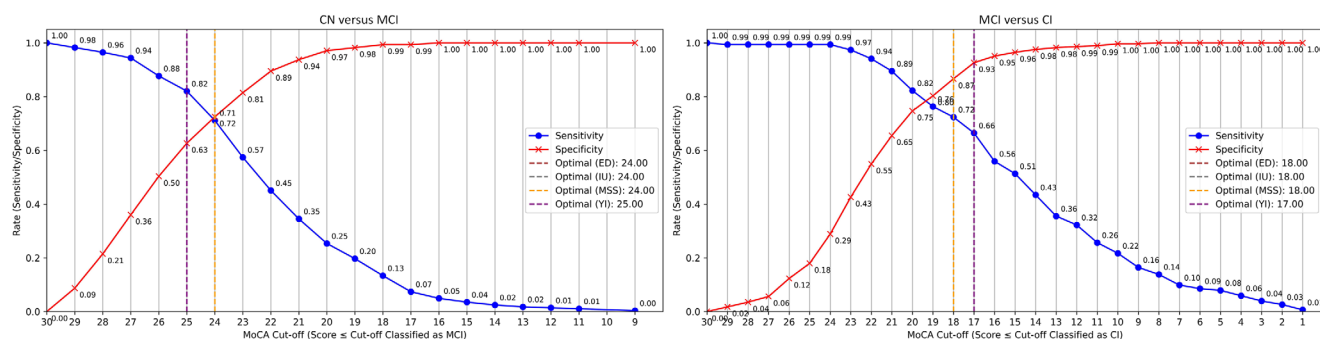
From the 961 initial eligible participants, 78 were excluded due to a lack of one or more neurocognitive tests. A total of 883 participants from the ADNI cohort were included in the analysis. Several demographic and clinical characteristics differed significantly across diagnostic categories. Mean age increased with diagnostic severity: CN participants had a mean age of 77.1 years, participants with MCI had a mean age of 78.5 years, and CI participants had a mean age of 79.8 years ($p < 0.001$). Educational attainment also differed by group: mean years of education were 17.0 (CN), 16.2 (MCI), and 16.1 (CI) ($p < 0.001$). Although the overall sample included nearly equal numbers of males and females (441 males, 442 females), the distribution of sex varied by diagnostic group ($p < 0.001$); specifically, the proportion of male participants was higher in the MCI and CI groups compared with CN. Occupational category distributions also differed across groups ($p < 0.001$), with a higher proportion of CN and MCI participants reporting professional/executive occupations relative to CI participants. Family history of dementia in a sibling showed a small but statistically significant difference across groups ($p < 0.001$), with more reporting sibling history in the MCI and CI groups compared with CN. Finally, indices summarizing the chronicity and severity of non-cognitive medical illnesses were greater in the CI group than in CN and MCI ($p < 0.001$), indicating a higher overall burden of comorbid disease among participants with greater cognitive impairment.

TABLE 1 Cut-off values of various statistical methods for CN versus MCI.

Cognitive test	Cut-off method	Cut-off value	Sensitivity (%)	Specificity (%)	AUC
CDR	ED	0.5	85.92	90.60	0.88
	IU	0.5	85.92	90.60	0.88
	MSS	0.5	85.92	90.60	0.88
	YI	0.5	85.92	90.60	0.88
MMSE	ED	28	63.03	70.02	0.70
	IU	28	63.03	70.02	0.70
	MSS	28	63.03	70.02	0.70
	YI	28	63.03	70.02	0.70
MoCA	ED	24	71.13	72.48	0.79
	IU	24	71.13	72.48	0.79
	MSS	24	71.13	72.48	0.79
	YI	25	82.04	62.64	0.79

Abbreviations: AUC, area under the curve; CDR, Clinical Dementia Rating; CN, cognitively normal; ED, Euclidean distance; IU, Index of Union; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; MSS, maximum product of sensitivity and specificity; YI, Youden's Index.

Sensitivity and Specificity versus MoCA Cut-off

**FIGURE 1** Sensitivity and specificity for MoCA for CN versus MCI (left) and MCI versus CI (right) with all four statistical methods' optimal cut-offs.

The three cognitive assessments differed among groups. The mean MoCA were 26.1 (CN), 22.5 (MCI), and 14.9 (CI). The mean MMSE were 28.8 (CN), 27.3 (MCI), and 20.5 (CI). Lastly, the mean CDR were 0.0 (CN), 0.5 (MCI), and 1 (CI).

3.1 | CN versus MCI cut-off values (Table 1, Figures 1–3)

For the CDR, all four methods had an equal cut-off value of 0.5, with a sensitivity of 86%, specificity of 91%, and an AUC of 0.88 (Supplemental Figure (A)). All methods had the same cut-off value for MMSE of 28 (sensitivity 63%, specificity 70%, AUC 0.70) (Supplemental Figure (B)). For MoCA, YI had a higher cut-off value of 25 (82% sensitivity, 63% specificity, 0.79 AUC) (Supplemental Figure (C)), where ED, IU, and MSS had a cut-off value of 24 (71% sensitivity, 73% specificity, 0.79 AUC) (Supplemental Figure (C)).

3.2 | MCI versus CI cut-off values (Table 2, Figures 1–3)

For the CDR, all methods (ED, IU, MSS, and YI) had a cut-off value of 1 (76% sensitivity, 95% specificity, and 0.87 AUC) (Supplemental Figure (D)). All methods had the same cut-off value for a MMSE score of 24 (sensitivity 82%, specificity 87%, and 0.91 AUC) (Supplemental Figure (E)). For MoCA, YI had a lower cut-off value of 17 (67% sensitivity, 93% specificity, and 0.88 AUC) (Supplemental Figure (F)), where ED, IU, and MSS had a cut-off value of 18 (72% sensitivity, 87% specificity, and 0.88 AUC) (Supplemental Figure (F)).

4 | DISCUSSION

This cross-sectional study systematically evaluated four distinct statistical methodologies for determining optimal diagnostic thresholds

Sensitivity and Specificity versus MMSE Cut-off

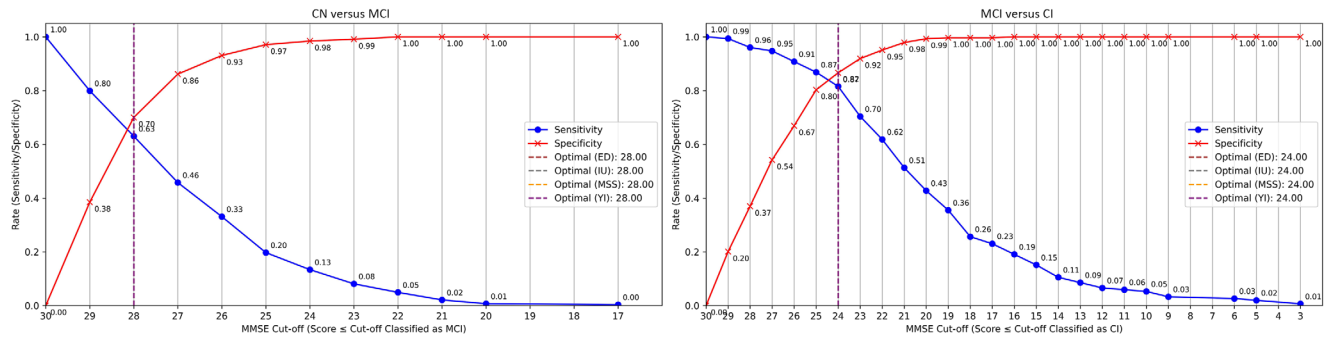


FIGURE 2 Sensitivity and specificity for the MMSE for CN versus MCI (left) and MCI versus CI (right) with all four statistical methods' optimal cut-offs.

Sensitivity and Specificity versus CDR Cut-off

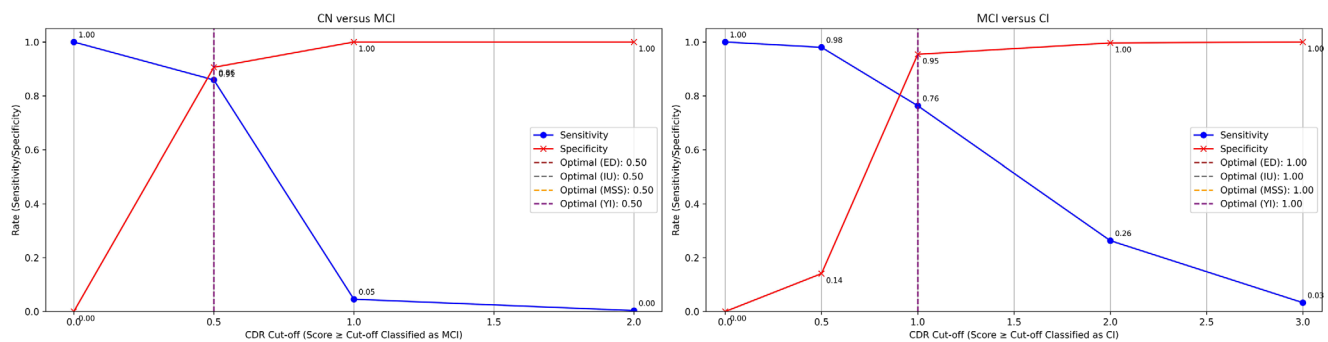


FIGURE 3 Sensitivity and specificity for the CDR for CN versus MCI (left) and MCI versus CI (right) with all four statistical methods' optimal cut-offs.

for the MoCA, the MMSE, and the CDR across the cognitive continuum using well-characterized ADNI participants.¹⁰ Our findings demonstrate methodological convergence across approaches for most clinical scenarios while highlighting important instrument-specific patterns that have significant implications for clinical decision-making and healthcare implementation.⁸

4.1 | Clinical translation and diagnostic performance

The demographic and clinical differences we observed across diagnostic groups align with established risk patterns for cognitive decline and have practical implications for screening and test interpretation. Older age and lower educational attainment were associated with greater diagnostic severity in our cohort; both are well-known correlates of cognitive test performance and can lower baseline scores or reduce the sensitivity of screening instruments.¹¹ A higher burden of chronic medical illness in the CI group may reflect increased vulnerability to cognitive decline or may contribute to functional impairment that influences clinician judgment. Differences in sex distribution across groups could arise from sampling variation, true

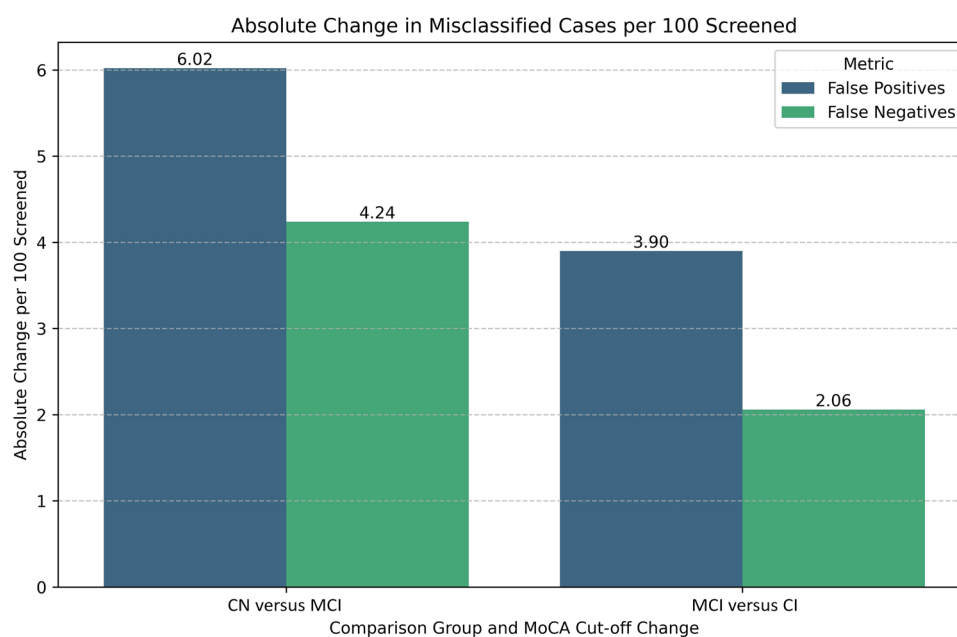
differences in disease prevalence or presentation by sex, or from differential help-seeking and referral patterns; these possibilities merit further study. Because age, education, comorbidity burden, and sex can all affect test performance, clinicians and researchers should account for these factors when selecting cut-offs and interpreting screening results.

Our results offer practical guidance for threshold selection that balances statistical optimization with clinical priorities. For MCI screening with the MoCA, a cut-off of 25 favors sensitivity (82% vs 71%), while a cut-off of 24 yields better balanced accuracy (specificity 73% vs 63%); raising the cut-off from 24 to 25 produces an absolute increase of 6.02 false positives and an absolute decrease of 4.24 false negatives per 100 screened (Figure 4). For distinguishing CI from MCI, the YI-derived MoCA cut-off of 17 increases specificity relative to an 18 threshold (93% vs 87%) at the cost of some sensitivity (67% vs 72%); moving from 17 to 18 yields an absolute increase of 3.90 false positives and an absolute decrease of 2.06 false negatives per 100 screened (Figure 4). These trade-offs suggest that current screening paradigms may under-represent the full severity spectrum of impairment, and that threshold choice should be guided by the clinical context.¹² By contrast, MMSE and CDR thresholds were stable across methods – MMSE: 28 (CN vs MCI) and 24 (MCI vs CI); CDR: 0.5 (CN vs MCI), and 1 (MCI vs CI) – sup-

TABLE 2 Cut-off values of various statistical methods for MCI versus CI.

Cognitive test	Cut-off method	Cut-off value	Sensitivity (%)	Specificity (%)	AUC
CDR	ED	1	76.32	95.42	0.87
	IU	1	76.32	95.42	0.87
	MSS	1	76.32	95.42	0.87
	YI	1	76.32	95.42	0.87
MMSE	ED	24	81.58	86.62	0.91
	IU	24	81.58	86.62	0.91
	MSS	24	81.58	86.62	0.91
	YI	24	81.58	86.62	0.91
MoCA	ED	18	72.37	86.62	0.88
	IU	18	72.37	86.62	0.88
	MSS	18	72.37	86.62	0.88
	YI	17	66.45	92.61	0.88

Abbreviations: AUC, area under the curve; CDR, Clinical Dementia Rating; CI, clinical impairment; CN, cognitively normal; ED, Euclidean distance; IU, Index of Union; MCI, mild cognitive impairment; MMSE, Mini-Mental State Examination; MoCA, Montreal Cognitive Assessment; MSS, maximum product of sensitivity and specificity; YI, Youden's Index.

**FIGURE 4** Absolute change in misclassified cases per 100 screened following MoCA cut-off adjustments.

porting their relative robustness despite known limitations in detecting early decline.

4.2 | Population-specific considerations and external validity

A fundamental limitation affecting the generalizability of our findings is the demographic homogeneity of the ADNI cohort, which substantially overrepresents highly educated, White participants compared

to general population demographics.¹⁰ ADNI's historical underrepresentation of ethnoculturally diverse populations (11% vs 25% in census data for adults 60+) and individuals with lower educational attainment (15% vs expected higher proportions in clinical populations) raises critical questions about threshold validity across diverse populations.^{13,14}

Extensive research demonstrates significant population-specific variation in optimal cognitive screening thresholds.^{12,14–16} MoCA cutoffs for MCI detection vary systematically by race/ethnicity and educational attainment, with required threshold reductions as edu-

cation decreases.^{12,14–16} Our mean educational attainment of 16.6 years exceeds that of many clinical populations, potentially limiting applicability to less educated groups who demonstrate systematically lower performance on cognitive screening instruments independent of pathological processes.¹⁵

4.3 | Implementation challenges and healthcare system considerations

Translation of optimized thresholds into clinical practice faces substantial implementation barriers that must be addressed.¹⁵ The CDR's superior diagnostic performance comes at significant resource cost, requiring trained clinicians and extended administration time that may be prohibitive in many healthcare settings.⁴ The recent development of electronic CDR (eCDR) versions shows promise for addressing scalability limitations while maintaining diagnostic accuracy.¹⁷

Given that most patients are seen by primary care specialists, early onset could be diagnosed in these settings.^{18,19} However, primary care implementation faces multiple barriers, including insufficient time for comprehensive cognitive assessment, limited training in threshold selection, and inadequate infrastructure for specialized cognitive evaluation.^{18,19} Studies document that primary care providers report time constraints, inadequate training in cognitive assessment, and lack of clear guidelines for threshold selection as primary barriers to effective cognitive screening implementation.^{18,19}

The population-specific nature of optimal thresholds creates additional implementation complexity, requiring clinical decision-support tools that can guide threshold selection based on patient demographic characteristics.^{12–15,18} This necessitates comprehensive provider education and systematic approaches to ensuring equitable diagnostic accuracy across diverse populations.^{18,19}

4.4 | Limitations and future research priorities

A central limitation of this study is that the reference standard is a clinician diagnosis derived from the synthesis of multiple data sources rather than an independent, biomarker-confirmed classification.^{10,20} ADNI site physicians routinely consider cognitive test results alongside clinical history, informant reports, functional assessments, imaging, and laboratory data when assigning diagnostic labels, and ADNI enrollment criteria permit overlapping score ranges across diagnostic groups. These features create the potential for incorporation bias: Cognitive instruments that are more heavily used in routine clinical assessment may correlate more closely with the physician label and therefore appear to perform better. Future validation of the proposed cut-offs should prioritize independent reference standards – such as biomarker-confirmed diagnoses or blinded adjudication panels – and targeted analyses designed to quantify and adjust for incorporation bias.^{10,20}

In conclusion, our comprehensive evaluation provides evidence-based thresholds for three widely used cognitive screening instru-

ments while highlighting the critical importance of cut-off selection and population-specific validation^{5,7} for the MoCA. The MMSE and the CDR cut-off consistency supports the established literature. Future research focusing on diverse population validation, and integration with emerging biomarker frameworks will be necessary for translating these findings into improved clinical care for individuals across the cognitive spectrum.^{10,20}

ACKNOWLEDGMENTS

Data collection and sharing for the Alzheimer's Disease Neuroimaging Initiative (ADNI) is funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organization is the Northern California Institute for Research and Education. In the past, ADNI has also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of Health Research, and private sector contributions through the Foundation for the National Institutes of Health (FNIH) including generous contributions from the following: AbbVie, Alzheimer's Association; Alzheimer's Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd. and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC; Johnson & Johnson Pharmaceutical Research & Development LLC; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

Consent was not necessary.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Atiya S, Aschebrook-Kilfoy B, Konda S; Alzheimer's Disease Neuroimaging Initiative. Comparative methods for determining MoCA, MMSE, and CDR cut-offs in Alzheimer's disease: insights from the ADNI study. *Alzheimer's Dement*. 2026;18:e70291. <https://doi.org/10.1002/dad2.70291>